

A Multimodal AI Agent Framework for Autism Behavioral Trait Detection in Children

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**A Capstone Project Report submitted in partial fulfillment of the requirements
for the degree of Bachelor of Science in Computer Science and Engineering**



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April, 2026

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Declaration

We hereby declare that this capstone project report entitled "A Multimodal AI Agent Framework for Autism Behavioral Trait Detection in Children" is our original work and has been carried out under the supervision of **Md AI Imran**, Senior Lecturer, Department of Computer Science and Engineering, East West University, Dhaka, Bangladesh. We further declare that this work has not been submitted anywhere else for the award of any degree, diploma, or other academic qualification.

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Letter of Acceptance

The capstone project report entitled "A Multimodal AI Agent Framework for Autism Behavioral Trait Detection in Children" is submitted by Sharafat Ahmed, Lima Prodhan, Rajon Das, and Jannatul Ferdous to the Department of Computer Science and Engineering, East West University, Dhaka, Bangladesh and is accepted for the partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science and Engineering.

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Abstract

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition characterised by persistent deficits in social communication and restricted, repetitive patterns of behaviour. Despite the availability of validated screening instruments, early identification remains a global challenge due to limited access to specialists, high diagnostic costs, and the absence of scalable, caregiver-friendly tools. This project presents a Multimodal AI Agent Framework for Autism Trait Detection in Children, designed as an end-to-end conversational screening and support system.

The proposed framework integrates two independently trained machine learning models within a multi-agent orchestration architecture. A gradient-boosted decision tree model (XGBoost) is employed for structured questionnaire-based screening using Q-CHAT-10 features, achieving 98.6% test accuracy and near-perfect discriminative performance. A fine-tuned transformer-based model (BERT) is used for free-text behavioural analysis, achieving 88% accuracy and strong generalisation under data-constrained conditions through semantically conservative augmentation techniques.

Beyond classification, the system incorporates a DSM-5-aligned behavioural trait taxonomy, a concern-level scoring mechanism, and companion support features including personalised social story generation and visual schedule creation. A strict architectural boundary ensures that all diagnostic outputs are generated exclusively by validated machine learning models, while the language model is restricted to explanation, routing, and caregiver interaction.

The dual-modality design significantly enhances accessibility by accommodating both structured and narrative caregiver inputs. The proposed system demonstrates the potential of combining machine learning and agent-based AI to deliver scalable, low-cost, and accessible early ASD screening and support tools, particularly in resource-constrained settings.

Acknowledgment

Completing this thesis has been a meaningful academic journey made possible through the guidance, support, and encouragement of many individuals. It is difficult to fully express the depth of gratitude owed to those who contributed, directly and indirectly, to the successful completion of this work.

First and foremost, I would like to express my sincere appreciation to my respected supervisor, **Md Al Imran**, whose continuous guidance, insightful feedback, and patient mentorship were instrumental throughout the entire research process. His thoughtful suggestions helped shape the direction of this work, refine its technical depth, and maintain a strong academic standard. His support and encouragement played a crucial role in overcoming challenges and completing this thesis successfully.

I am also grateful to the Department of Computer Science and Engineering at East West University, Bangladesh, for providing a supportive academic environment and the necessary resources to carry out this research. The knowledge and skills gained throughout my academic journey have been fundamental in the development of this project. I would like to acknowledge my peers and well-wishers whose discussions, feedback, and encouragement contributed to the progress of this work. Their support, whether through technical insights or motivation during challenging phases, has been deeply valuable.

Finally, I extend my heartfelt gratitude to my family for their unwavering support, patience, and understanding throughout this journey. Their encouragement has been a constant source of strength and motivation. This thesis represents not only an academic accomplishment but also the collective influence of everyone who has supported me along the way. I remain sincerely grateful to all who have contributed to this journey.

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Chapter 1: Introduction

1.1 Background, Overview, and Motivation

Autism Spectrum Disorder (ASD) is a heterogeneous neurodevelopmental condition characterised by persistent deficits in social communication and interaction, alongside restricted and repetitive patterns of behaviour [1]. The global prevalence of ASD has increased significantly over recent decades, with current epidemiological estimates indicating approximately one in thirty-six children being affected. This rising prevalence has intensified the demand for early identification and intervention.

Early detection of ASD is critically important due to the high neuroplasticity of the developing brain during early childhood. Research consistently demonstrates that interventions initiated before the age of three lead to substantially improved cognitive, behavioural, and social outcomes compared to later interventions [1]. However, despite the availability of validated screening instruments, the median age of diagnosis remains significantly delayed.

This delay is particularly pronounced in low- and middle-income regions, where access to specialised paediatric and psychological services is limited. Structural barriers such as the shortage of trained professionals, high costs of clinical assessment, and geographic inaccessibility contribute to a substantial gap between initial parental concern and formal diagnosis. Caregivers often observe early atypical behaviours but lack accessible tools to translate these observations into actionable clinical insights.

Recent advances in artificial intelligence, particularly in machine learning and natural language processing, offer promising opportunities to address these challenges. Machine learning models have demonstrated strong performance in structured ASD screening tasks [5], while transformer-based language models have enabled the analysis of unstructured clinical text with increasing accuracy [2]. At the same time, agent-based AI systems have emerged as a powerful paradigm for building interactive, stateful applications capable of handling complex user interactions.

This project is motivated by the convergence of these technological developments. A conversational AI system that integrates validated screening models with natural language interaction has the potential to provide scalable, low-cost, and accessible support to caregivers. Such a system can not only perform screening but also deliver personalised guidance and supportive resources, thereby extending its utility beyond traditional diagnostic tools.

1.2 Problem Statement and Analysis

Existing approaches to ASD screening can be broadly categorised into three groups, each with inherent limitations.

First, formal clinical assessments conducted by developmental paediatricians or clinical psychologists represent the gold standard for diagnosis. However, these assessments are resource-intensive, time-consuming, and inaccessible to a large portion of the global population.

Second, validated screening instruments such as Q-CHAT-10 and M-CHAT-R/F provide structured approaches to early detection. While these tools are effective, they are typically administered in clinical settings and assume that caregivers can accurately respond to predefined binary or Likert-scale questions. This assumption does not always hold in real-world contexts, where caregivers may find it easier to describe behaviours narratively rather than map them to structured formats.

Third, emerging digital screening tools have been proposed in research settings, often leveraging machine learning techniques. However, most existing systems are limited to a single input modality and lack conversational capabilities. They typically function as static tools that terminate after producing a classification result, without offering explanation, follow-up interaction, or ongoing support.

A key gap identified in the literature is the absence of integrated systems that support both structured questionnaire inputs and unstructured behavioural descriptions within a single framework. Additionally, existing systems rarely incorporate mechanisms to prevent unreliable outputs from language models, despite documented concerns regarding hallucination in clinical AI applications [2].

Another limitation is the lack of continuity in user interaction. Most tools provide a one-time result without enabling caregivers to ask follow-up questions, understand trait-level interpretations, or access supportive interventions such as behavioural guidance or structured routines.

This project addresses these gaps by proposing a multimodal, conversational AI framework that integrates multiple machine learning models within a controlled agent-based architecture, ensuring both flexibility of input and reliability of output.

1.3 Project Objectives

The primary objectives of this project are as follows:

- To develop and evaluate a machine learning model based on XGBoost for structured ASD screening using Q-CHAT-10 features.
- To design and fine-tune a transformer-based model (BERT) for classifying free-text behavioural descriptions of children.
- To implement a multimodal framework that supports both structured and unstructured input modalities within a unified system.
- To design a multi-agent orchestration architecture using a graph-based framework to manage conversational flow and system state.
- To establish a strict architectural boundary ensuring that all diagnostic outputs are generated exclusively by machine learning models.
- To develop a behavioural trait taxonomy aligned with DSM-5 criteria and implement a corresponding concern-level scoring mechanism.
- To integrate companion support features, including social story generation and visual schedule creation, to enhance system utility.
- To design a user-friendly conversational interface accessible to non-specialist caregivers.

1.4 Project Contributions

The key contributions of this work can be summarised as follows:

Contribution 1 — Multimodal ASD screening framework:

A unified system capable of processing both structured questionnaire data and free-text behavioural descriptions, enabling broader accessibility across diverse user groups.

Contribution 2 — Trustworthy AI architecture with anti-hallucination boundary:

A system design that explicitly separates machine learning-based diagnostic outputs from language model-generated content, addressing known reliability concerns in AI-assisted healthcare systems [2].

Contribution 3 — Behavioural trait taxonomy with clinical grounding:

A structured mapping of behavioural features to DSM-5-aligned categories, providing interpretable and clinically meaningful outputs.

Contribution 4 — Data-efficient transformer fine-tuning strategy:

A method for improving performance on small text datasets using semantically controlled augmentation techniques, preserving domain-specific meaning while increasing data diversity.

Contribution 5 — Integration of companion support capabilities:

The incorporation of personalised social stories, visual schedules, and contextual guidance, transforming the system from a screening tool into an ongoing support platform.

1.5 Project Outline

The remainder of this report is structured as follows:

- **Chapter 2** reviews existing literature on machine learning approaches for ASD screening, natural language processing in healthcare, and agent-based AI systems.
- **Chapter 3** describes the methodology, including datasets, preprocessing, model architectures, augmentation strategies, and system design.
- **Chapter 4** presents experimental results, performance evaluation, and detailed analysis of both classification models and the overall system.
- **Chapter 5** concludes the report by summarising contributions, discussing limitations, and outlining directions for future research, along with a brief business model analysis.

Chapter 2: Related Work

2.1 Survey of the State-of-the-Art

2.1.1 Machine Learning for ASD Screening

The application of machine learning techniques to Autism Spectrum Disorder (ASD) screening has evolved significantly over the past decade, driven by the increasing availability of behavioural datasets and the need for scalable diagnostic support tools. Early studies primarily focused on classical machine learning algorithms such as decision trees, k-nearest neighbours (KNN), naïve Bayes, and support vector machines. A comprehensive review by Fadi Thabtah highlighted that these models achieved promising classification performance across multiple ASD screening instruments, particularly when applied to structured questionnaire-based datasets [5]. However, the review also identified critical methodological limitations in prior work, including the widespread issue of data leakage, where derived features such as total questionnaire scores were inadvertently included in training, artificially inflating model performance.

The development of validated screening instruments has played a crucial role in enabling machine learning applications. The Q-CHAT-10, introduced by Catherine Allison, Bonnie Auyeung, and Simon Baron-Cohen, represents a concise and clinically validated tool for early ASD detection in toddlers [3]. Its binary response structure and focus on core social-communication behaviours make it particularly suitable for computational modelling. Empirical studies have demonstrated that specific items within the Q-CHAT-10, such as response to name, eye contact, and joint attention behaviours, exhibit strong discriminative power, which aligns with established clinical understanding of early ASD markers.

In recent years, ensemble learning methods, particularly gradient boosting algorithms, have emerged as the dominant approach for tabular ASD datasets. The introduction of XGBoost by Tianqi Chen and Carlos Guestrin has significantly advanced performance in structured data classification tasks [10]. XGBoost's ability to model complex feature interactions through sequential residual learning, combined with built-in regularisation mechanisms, makes it especially effective for behavioural datasets where feature dependencies are non-linear. Recent studies consistently report that gradient-boosted models outperform traditional baselines such as logistic regression and random forests in ASD screening tasks, particularly when careful attention is paid to class imbalance and feature selection.

Despite these advances, an important limitation remains: most existing machine learning approaches rely exclusively on structured questionnaire inputs. This assumption restricts usability in real-world contexts, where caregivers may not be able to accurately interpret or respond to predefined questions. Consequently, there is a growing need for systems that can incorporate more flexible input modalities while maintaining robust classification performance.

2.1.2 Natural Language Processing in Healthcare

The rapid development of transformer-based language models has significantly expanded the role of natural language processing (NLP) in healthcare applications. The introduction of BERT by Jacob Devlin et al. marked a major shift in NLP by enabling bidirectional contextual understanding of text [6]. Unlike earlier models that processed text sequentially, BERT captures contextual dependencies across entire sequences, making it particularly effective for classification tasks involving nuanced linguistic patterns.

In the medical domain, transformer-based models have been applied to a wide range of tasks, including clinical note classification, disease prediction, and patient communication analysis. A recent review by Ajay J Thirunavukarasu et al. highlights that large language models can achieve performance comparable to domain experts in certain diagnostic and reasoning tasks, while also emphasising critical challenges

such as hallucination and reliability [2]. Hallucination, defined as the generation of plausible but factually incorrect information, represents a significant risk in clinical contexts where accuracy is paramount.

For ASD-specific applications, the use of NLP remains relatively underexplored due to the scarcity of labelled text datasets. The ASD Behavioural Text Dataset developed by Fadi Thabtah provides one of the few publicly available corpora for this task [7]. However, its limited size presents a substantial challenge for training deep learning models, particularly large transformer architectures that typically require extensive data.

To address data scarcity, data augmentation techniques have been widely adopted in NLP research. The EDA (Easy Data Augmentation) framework proposed by Jason Wei and Kai Zou demonstrated that simple operations such as synonym replacement, random insertion, and word swapping can significantly improve classification performance in low-resource settings [4]. More recent approaches have extended these techniques by leveraging contextual language models to generate semantically coherent augmentations, thereby preserving the underlying meaning of the text while introducing variability.

Despite these advances, the application of NLP to ASD screening remains limited by dataset size and variability in caregiver language. This highlights the importance of designing augmentation strategies that maintain clinical relevance while enabling effective model training.

2.1.3 Multi-Agent and Agentic AI Systems in Healthcare

The emergence of agent-based AI systems represents a significant paradigm shift in the design of intelligent applications. Unlike traditional monolithic models, agent-based architectures decompose complex tasks into smaller, specialised components that interact through defined workflows. This approach is particularly well-suited to healthcare applications, where tasks such as data collection, classification, explanation, and user interaction must be carefully coordinated.

Graph-based orchestration frameworks, such as LangGraph, enable the implementation of stateful multi-agent systems with explicit control over information flow and decision-making processes. These frameworks support persistent memory, conditional routing, and modular agent design, allowing developers to construct systems that can handle multi-turn interactions and dynamic user inputs. In the context of ASD screening, such capabilities are essential for replicating the iterative nature of clinical interviews and caregiver consultations.

A key challenge in integrating large language models into healthcare systems is ensuring reliability and preventing the generation of misleading or incorrect information. Recent research has emphasised the need for hybrid architectures that combine symbolic control mechanisms with statistical learning models to improve trustworthiness. The concept of an “anti-hallucination boundary,” where language models are restricted from generating critical outputs such as diagnoses, aligns with emerging best practices in trustworthy AI system design.

In addition to screening, there is growing interest in leveraging AI for therapeutic and assistive applications. Social stories, originally introduced by Carol Gray, have been widely used as an intervention strategy to improve social understanding and reduce anxiety in children with ASD [8]. Similarly, visual schedules have been shown to support behavioural regulation and transition management in educational and home environments [9]. The integration of these evidence-based interventions into AI systems represents a natural extension of their traditional use, enabling scalable and personalised delivery.

Furthermore, retrieval-based methods using vector databases such as FAISS, developed by Jeff Johnson and colleagues, have enabled efficient semantic search over large collections of documents [11]. When combined with sentence embedding models such as Sentence-BERT, these systems can retrieve contextually relevant materials with high accuracy [12]. This capability is particularly valuable for generating personalised support content, as it allows systems to leverage existing knowledge bases while adapting outputs to individual user needs.

2.2 Summary of Research Gap

The literature review reveals several important gaps that motivate the present work. First, most existing ASD screening systems are limited to a single input modality, typically structured questionnaires, which restricts accessibility for diverse user populations. Second, while NLP techniques have shown promise in healthcare applications, their use in ASD screening remains constrained by limited data availability and the absence of robust multimodal integration.

Third, current AI-based screening tools often lack mechanisms to ensure reliability, particularly when incorporating large language models. The risk of hallucination in clinical contexts necessitates architectural safeguards that are not commonly implemented in existing systems. Finally, most tools focus solely on classification, without providing ongoing support or intervention resources for caregivers.

This project addresses these gaps by proposing a multimodal, agent-based framework that integrates structured and unstructured inputs, enforces a strict

separation between diagnostic and conversational components, and extends functionality beyond screening to include personalised caregiver support.

Chapter 3: Methodology

3.1 Materials

3.1.1 Dataset Collection

The proposed framework is built upon two independently sourced datasets, each corresponding to one of the system’s two input modalities: structured questionnaire-based screening and free-text behavioural description analysis. This dual-dataset design reflects the core objective of enabling multimodal interaction within a unified screening system.

The first dataset is the Q-CHAT-10 Toddler Autism Screening Dataset, obtained from a publicly available repository on Kaggle. This dataset comprises 1,054 records representing completed screening assessments for children aged between 12 and 36 months. Each record includes ten binary features corresponding to the Q-CHAT-10 questionnaire items (A1–A10), along with additional demographic and contextual attributes such as sex, age in months, ethnicity, history of neonatal jaundice, family history of ASD, and the identity of the respondent. The target variable indicates the presence or absence of ASD traits. The structured and clinically grounded nature of this dataset makes it particularly suitable for supervised machine learning approaches.

The second dataset is the ASD Behavioural Text Dataset (Mendeley Data, version 2), which contains 332 free-text descriptions of child behaviour. Each entry is labelled as ASD or Non-ASD and annotated with one of fourteen behavioural categories. These textual descriptions represent natural caregiver or observer narratives, ranging from brief statements to longer behavioural accounts. Unlike the structured dataset, this corpus captures the variability and ambiguity inherent in real-world language, making it suitable for evaluating natural language processing models under low-resource conditions.

3.1.2 Dataset Exploration

A thorough exploratory data analysis was conducted to understand the statistical properties and quality of both datasets prior to model development.

For the Q-CHAT-10 dataset, the data matrix ($1,054 \times 18$) was examined for missing values and inconsistencies. No missing values were identified in the core feature columns, indicating high data completeness. The class distribution revealed a slight imbalance, with ASD-positive instances forming the majority class. Statistical hypothesis testing using the chi-square test ($\alpha = 0.05$) was performed to evaluate associations between categorical features and the target variable. The variable “sex” demonstrated a highly significant association with ASD traits ($p < 0.001$), consistent with established epidemiological findings of higher ASD prevalence in males. The presence of neonatal jaundice also showed statistical significance ($p = 0.020$), aligning with literature suggesting a potential link between early neurological risk factors and ASD.

Other variables, such as family history of ASD and respondent identity, did not exhibit statistically significant associations within this dataset. This may be attributed to limited subgroup representation and reduced statistical power. Ethnicity, while statistically significant, was excluded from further modelling due to concerns regarding bias and lack of causal interpretability.

To further quantify feature relationships, point-biserial correlation coefficients were computed between each feature and the binary target variable. The core questionnaire items (A1–A10) demonstrated moderate to strong correlations, confirming their discriminative value. In contrast, age exhibited only weak correlation, which is expected given the narrow developmental range of the sampled population.

For the ASD Behavioural Text Dataset, an initial data cleaning step identified and removed one erroneous record, resulting in a final dataset of 332 samples. The class distribution was found to be approximately balanced, eliminating the need for class reweighting during model training. Token length analysis, performed on the training split, indicated that 95% of samples contained fewer than 64 tokens after tokenisation. This observation informed the selection of a maximum sequence length of 64 tokens for the transformer model, ensuring efficient computation without significant information loss.

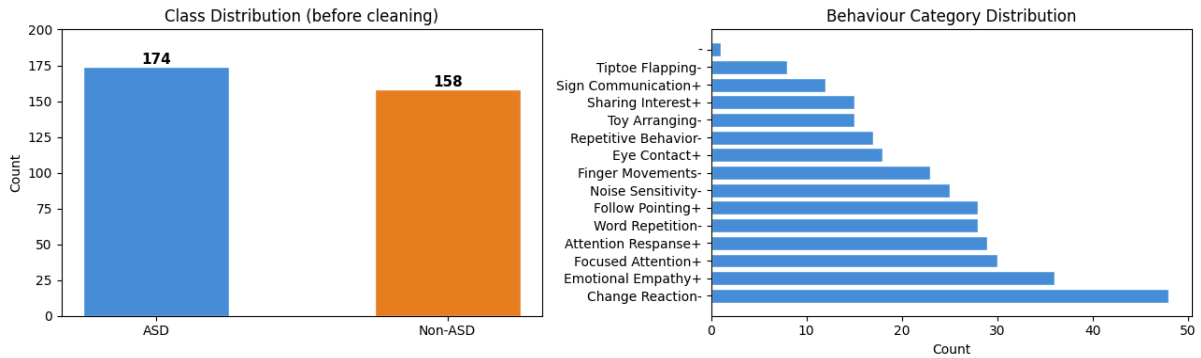


Figure 1 : class distribution and text column before cleaning

3.1.3 Dataset Sampling

To ensure robust model evaluation and prevent overfitting, appropriate sampling strategies were applied to both datasets.

For the Q-CHAT-10 dataset, a stratified 80/20 train-test split was implemented using standard machine learning practices. Stratification ensured that the class distribution was preserved across both training and test sets. The resulting partitions consisted of 843 training samples and 211 test samples. No explicit resampling techniques were applied; instead, class imbalance was addressed during model training through parameter tuning.

For the textual dataset, a three-way stratified split was adopted to facilitate model validation and early stopping. The dataset was divided into 70% training (232 samples), 15% validation (50 samples), and 15% test (50 samples). This approach ensured that model hyperparameters could be optimised using validation data while preserving the integrity of the test set for final performance evaluation.

3.1.4 Dataset Processing

Structured Dataset Preprocessing

The preprocessing pipeline for the structured dataset was designed to eliminate potential sources of bias and data leakage while preserving meaningful predictive features.

A critical step involved the removal of the “Qchat-10-Score” feature, which represents the sum of the questionnaire responses. Including this feature would introduce direct leakage, as it is a deterministic function of the input variables and

strongly correlated with the target. Additionally, non-informative identifiers such as “Case_No” were excluded from the feature set.

Categorical variables were encoded into binary representations to facilitate model compatibility. Specifically, sex, jaundice history, and family ASD history were converted into binary numerical values. The final feature vector consisted of eleven binary variables: the ten questionnaire items (A1–A10) and the sex attribute.

Given that all features were binary and on the same scale, no feature scaling was required. This is consistent with the properties of tree-based models such as XGBoost, which are invariant to feature scaling.

Text Dataset Augmentation

The relatively small size of the text dataset posed a significant challenge for training a deep learning model. To address this limitation, a two-stage data augmentation strategy was employed, applied exclusively to the training set to avoid data leakage.

The first stage involved synonym replacement using a lexical database. A predefined proportion of non-critical words within each text sample was replaced with semantically similar alternatives. To preserve domain-specific meaning, a protected vocabulary of clinically relevant terms was defined, ensuring that key concepts such as “eye contact” and “repetitive behaviour” remained unchanged.

The second stage utilised contextual augmentation based on a masked language model. In this approach, selected words were masked and replaced with contextually appropriate alternatives predicted by the model. Unlike simple lexical substitution, this method considers the surrounding context, resulting in more coherent and semantically consistent augmented samples.

The combination of these two strategies effectively tripled the size of the training dataset, increasing it from 232 to 696 samples. This augmentation process improved model generalisation while maintaining the integrity of the original semantic content

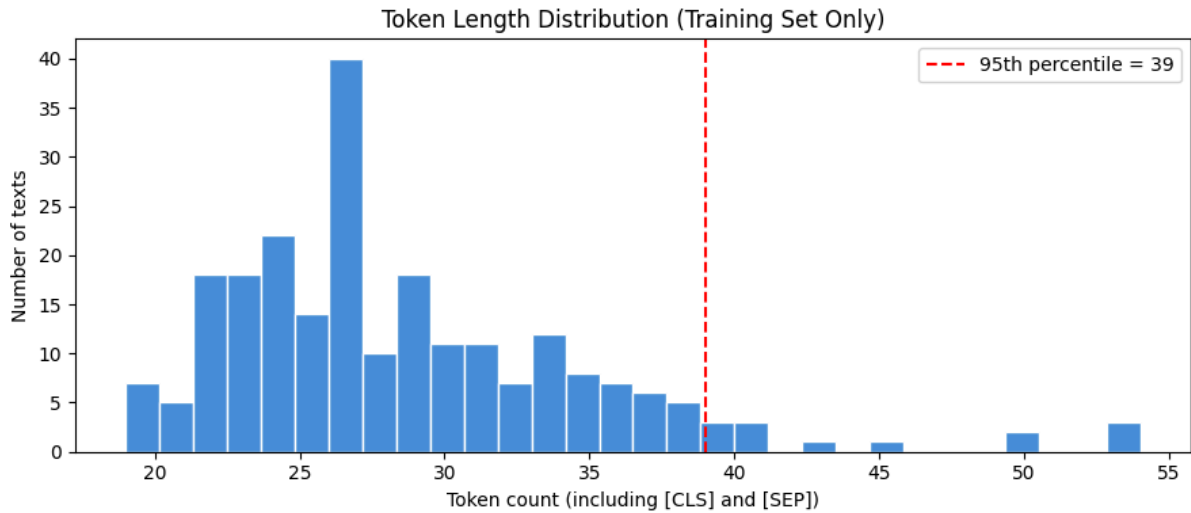


Figure 2 : Token Length

Tokenisation

All text samples were processed using a pre-trained tokenizer associated with the transformer model. Tokenisation included truncation and padding to a fixed maximum length, as well as the addition of special tokens required for classification tasks. Importantly, the tokenizer was applied consistently across training, validation, and test sets without additional fitting, ensuring methodological integrity.

3.1.5 Research Environment

All experiments were conducted in a cloud-based environment using GPU acceleration to support deep learning operations. The implementation utilised modern machine learning libraries, including PyTorch for model training, Hugging Face Transformers for NLP tasks, and scikit-learn for classical machine learning workflows. Additional tools such as FAISS were employed for efficient semantic retrieval, while Streamlit was used to develop the user interface.

3.2 Method

3.2.1 Model Design

The proposed system integrates two distinct machine learning models, each tailored to a specific input modality, within a unified architecture.

The first model is based on gradient boosting, a powerful ensemble technique for tabular data. XGBoost constructs an ensemble of decision trees in a sequential

manner, where each new tree learns to correct the residual errors of the previous ensemble. Regularisation techniques are incorporated to prevent overfitting, and class imbalance is handled through weighted loss functions.

The second model is a transformer-based neural network designed for text classification. BERT processes input text using self-attention mechanisms that capture contextual relationships between words. Fine-tuning adapts the pre-trained model to the specific classification task by updating all parameters jointly. The model outputs class probabilities, which are used to determine the final prediction.

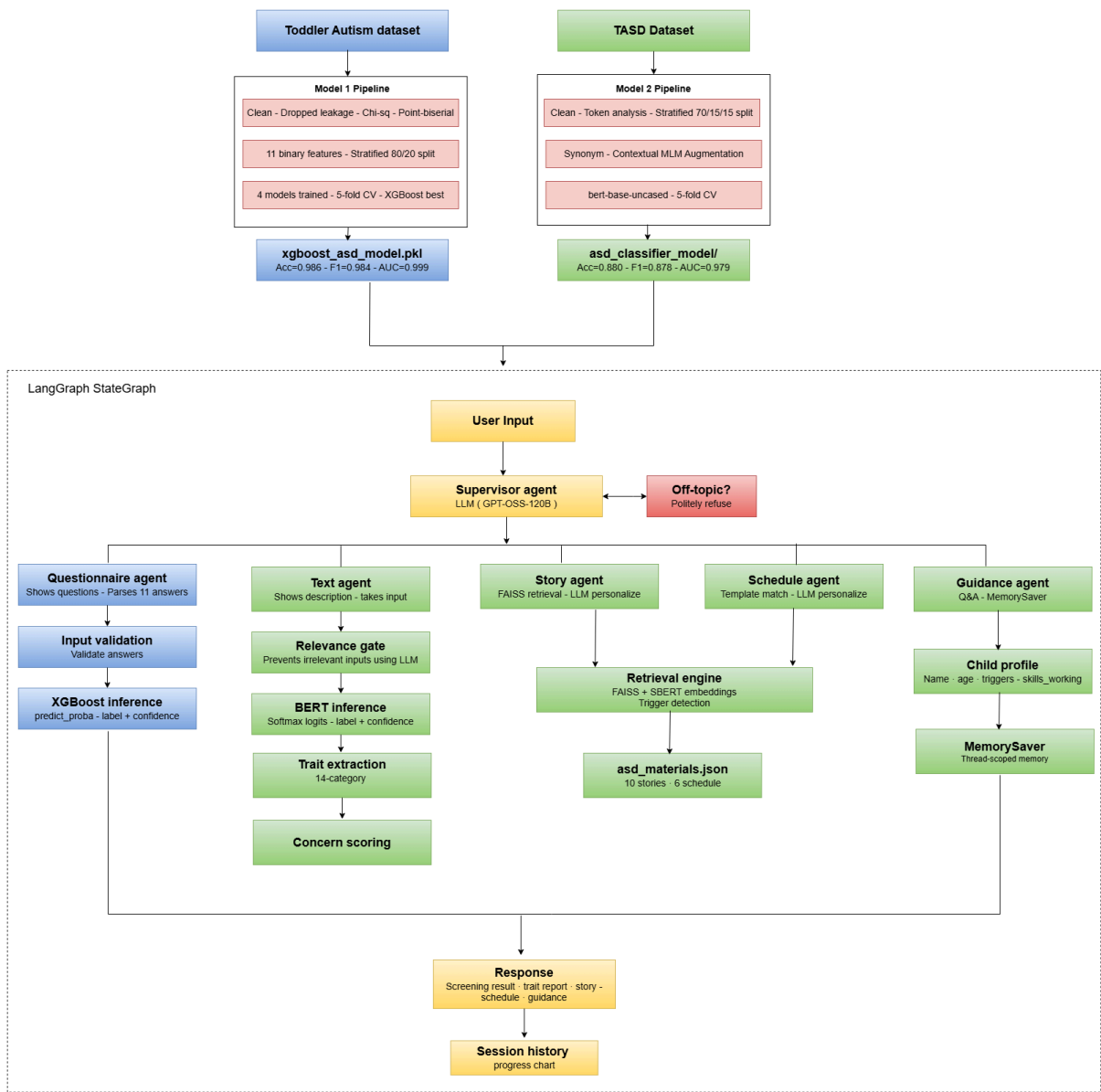


Figure 3 : Methodology

3.2.2 Multi-Agent Framework Design

The system architecture follows a multi-agent design pattern, where specialised components handle distinct tasks within the overall workflow. A central supervisory agent manages user interactions, classifies intents, and routes inputs to the appropriate processing modules.

Separate agents are responsible for handling structured questionnaire input and free-text descriptions. Additional agents manage supportive functionalities such as generating personalised content and retrieving relevant resources. This modular design enhances system flexibility, scalability, and maintainability.

A key design principle is the enforcement of a strict boundary between machine learning outputs and language model-generated responses. Diagnostic decisions are produced exclusively by trained classifiers, while the language model is restricted to explanation and interaction. This separation ensures reliability and mitigates risks associated with hallucination in generative models.

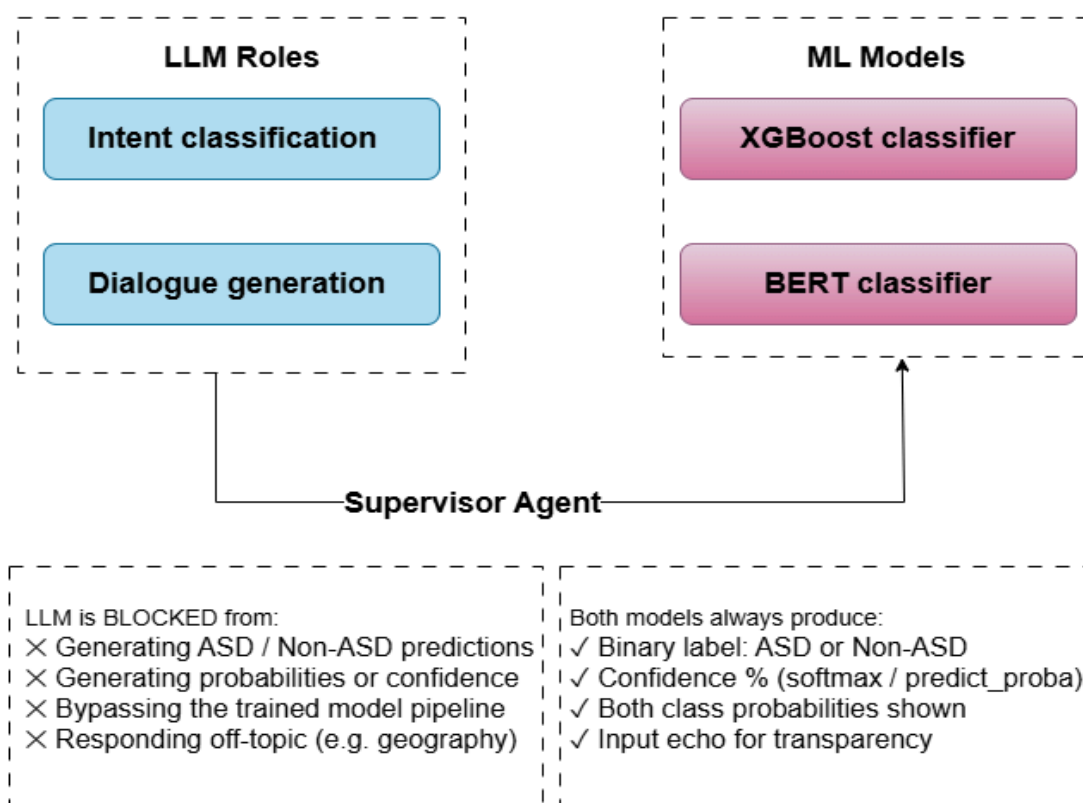


Figure 4 : What the LLM does vs what the ML models do

3.2.3 Experimental Setup

To evaluate model performance, multiple baseline algorithms were implemented alongside the proposed models. For the structured dataset, classifiers including Random Forest, KNN, and Logistic Regression were compared against XGBoost using standard evaluation metrics such as accuracy, precision, recall, and F1-score. Cross-validation was employed to assess generalisation.

For the text classification task, the transformer model was evaluated using a held-out test set and cross-validation. Performance metrics included accuracy, macro-averaged precision and recall, F1-score, and AUC-ROC. Confidence intervals were calculated to account for the limited size of the test set.

3.3 Summary

This chapter presented the complete methodological framework underlying the proposed system. The approach combines structured and unstructured data processing, robust preprocessing techniques, data augmentation strategies, and advanced machine learning models within a multi-agent architecture. Particular emphasis was placed on preventing data leakage, ensuring model reliability, and maintaining a clear separation between classification and conversational components.

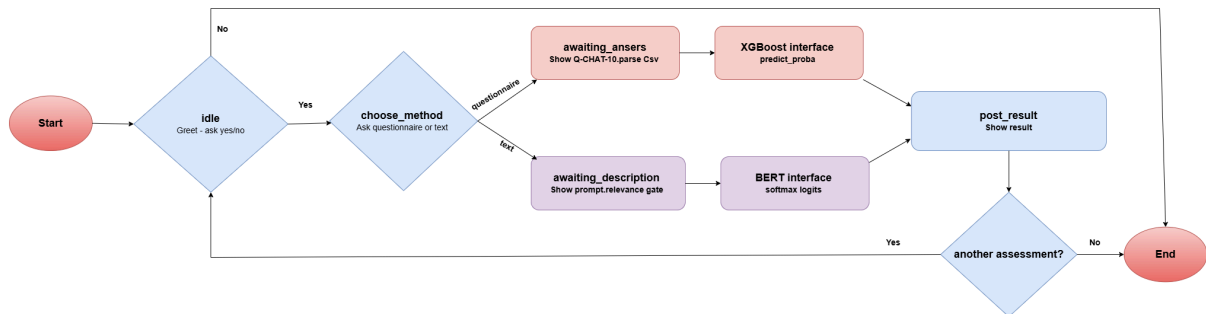


Figure 5 : ML Model prediction flow

Chapter 4: Results and Discussion

4.1 Obtained Results

4.1.1 XGBoost Classifier (Structured Modality)

The performance of the structured screening modality was evaluated using four machine learning classifiers: Random Forest, K-Nearest Neighbours (KNN), Logistic Regression, and XGBoost. All models were trained and evaluated on an identical stratified 80/20 train-test split to ensure fair comparison. Among these models, XGBoost demonstrated the highest performance across all evaluation metrics, achieving a test accuracy of 98.58%, along with consistently superior macro-averaged precision, recall, and F1-score.

To further assess generalisation, a five-fold stratified cross-validation procedure was conducted on the training set. XGBoost achieved a mean cross-validation accuracy of 98.58% with a very low standard deviation ($\pm 0.41\%$), indicating stable and consistent performance across different data partitions. This stability is particularly important in clinical screening contexts, where reliability across diverse samples is essential.

The model's discriminative ability was further evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), where XGBoost achieved a value of 0.9997. This near-perfect score indicates that the model can effectively distinguish between ASD-positive and ASD-negative cases across a wide range of decision thresholds.

However, it is important to interpret this exceptionally high performance with caution. The Q-CHAT-10 instrument itself is specifically designed to capture highly discriminative behavioural indicators of ASD, and its binary feature structure reduces ambiguity in model learning. As a result, strong separability in the feature space is expected. Additionally, careful preprocessing steps, including the removal of leakage-prone features, ensure that the observed performance reflects genuine predictive capability rather than artefacts of the dataset.

Feature importance analysis revealed that specific questionnaire items—particularly those related to response to name, eye contact, and social reciprocity—contributed most significantly to the model's predictions. This finding aligns closely with established clinical knowledge, reinforcing the interpretability and validity of the model.

Model	Accuracy	Precision	Recall	F1 (macro)	CV Accuracy
XGBoost	0.9858	0.9779	0.9897	0.9835	0.9763 $\pm$ 0.014
Random Forest	0.9621	0.9555	0.9555	0.9555	0.9597 $\pm$ 0.017
Logistic Reg.	0.9526	0.9413	0.9487	0.9449	0.9573 $\pm$ 0.026
KNN	0.9431	0.9259	0.9461	0.9349	0.9300 $\pm$ 0.030

Table 1 : Model Comparison

4.1.2 BERT Classifier (Free-Text Modality)

The transformer-based text classification model demonstrated strong performance given the constraints of a relatively small dataset. On the held-out test set of 50 samples, the model achieved an accuracy of 88%, with macro-averaged precision, recall, and F1-score also at 0.88. The AUC-ROC score of 0.96 indicates a high level of discriminative capability, suggesting that the model successfully captures meaningful patterns in caregiver-provided behavioural descriptions.

Due to the limited size of the test set, a Wilson score confidence interval was calculated, yielding a range of approximately 76.1% to 94.5%. This relatively wide interval reflects the inherent uncertainty associated with small sample sizes and highlights the importance of interpreting point estimates with caution.

To obtain a more robust estimate of generalisation performance, five-fold cross-validation was performed across the full dataset. The model achieved a mean accuracy of 88.25% with a standard deviation of 4.30%. While some variability across folds was observed, this is expected in low-resource settings where the composition of training subsets can significantly influence model behaviour.

The training process exhibited stable convergence, with training loss decreasing steadily and validation loss stabilising without signs of severe overfitting. Early stopping proved effective in preventing unnecessary training epochs and preserving generalisation performance.

The applied data augmentation strategy played a crucial role in enabling effective model training. By expanding the dataset through both synonym-based and contextual augmentation, the model was exposed to a wider range of linguistic variations while maintaining semantic consistency. The use of a protected vocabulary ensured that clinically relevant terms were preserved, preventing degradation of domain-specific meaning.

Metric	Score
Accuracy	0.8800 (95% CI: [0.762, 0.944])
Precision (macro)	0.8917
Recall (macro)	0.8766
F1 (macro)	0.8782
AUC-ROC	0.9792
5-Fold CV Accuracy	0.8825 $\pm$ 0.0423

Table 2 : BERT Results

4.2 In-Depth Results Analysis

4.2.1 Analysis of Structured Modality

The near-perfect performance of the XGBoost model can be attributed to both the quality of the dataset and the suitability of the algorithm. The Q-CHAT-10 features are inherently designed to capture key behavioural markers of ASD, resulting in a feature space that is highly separable. XGBoost's ability to model non-linear interactions and leverage feature importance weighting allows it to fully exploit this structure.

The observed correlations among key questionnaire items, particularly those related to joint attention and social engagement, reflect underlying behavioural constructs that are central to ASD diagnosis. The model effectively captures these relationships, contributing to its high predictive performance.

Importantly, the decision to exclude certain demographic features, such as ethnicity, reflects a deliberate effort to prioritise fairness and generalisability. While including such features may yield marginal performance improvements, it risks introducing bias and reducing the ethical robustness of the system.

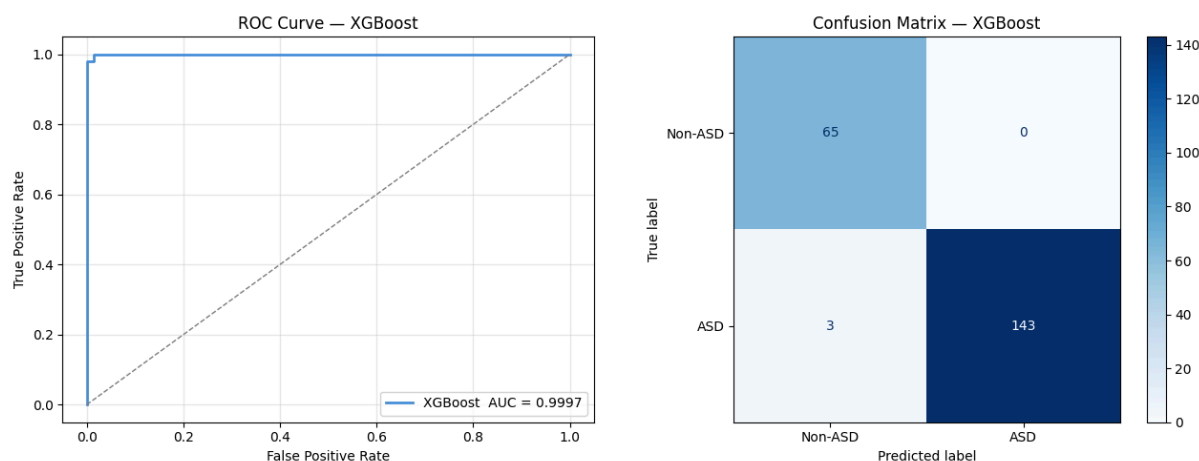


Figure 6: XGBoost ROC curve, confusion matrix

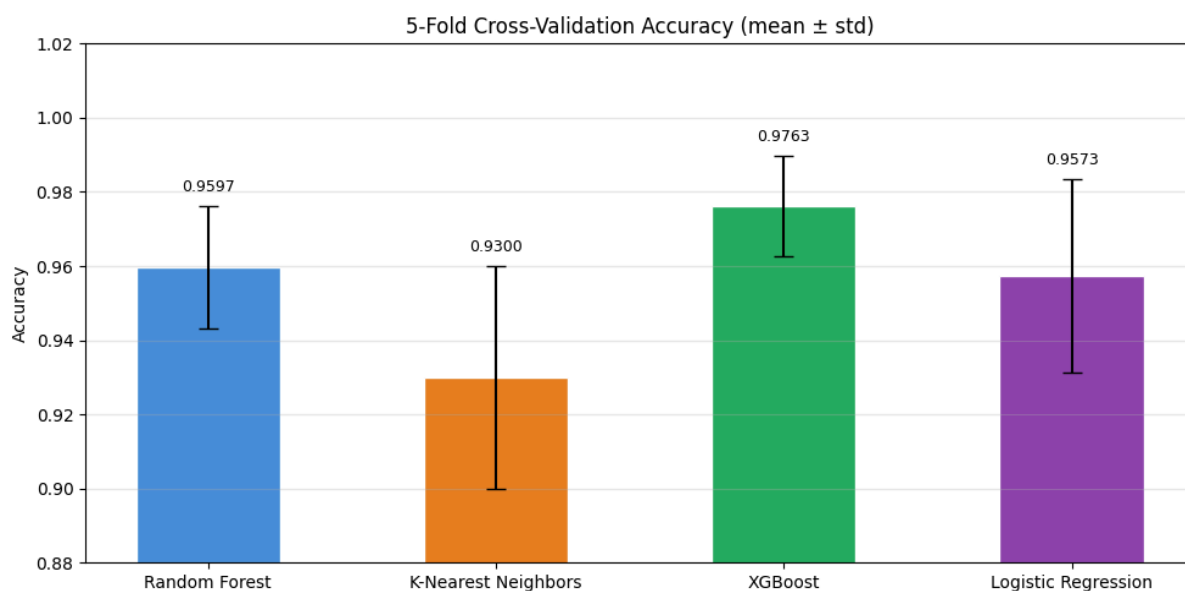


Figure 7: 5-fold CV results

4.2.2 Analysis of Free-Text Modality

The performance of the BERT-based model demonstrates that unstructured behavioural descriptions contain sufficient signal for meaningful classification, even in low-resource scenarios. The model's ability to generalise across different textual expressions suggests that it successfully captures semantic patterns associated with ASD-related behaviours.

Nevertheless, the variability observed across cross-validation folds highlights the sensitivity of deep learning models to dataset size and composition. This underscores the importance of cautious interpretation and the need for larger, more diverse datasets in future work.

The augmentation strategy proved to be a critical component of the methodology. By combining lexical and contextual augmentation techniques, the dataset was expanded without introducing significant noise. This approach strikes a balance between increasing data diversity and preserving semantic integrity, which is particularly important in clinical applications.

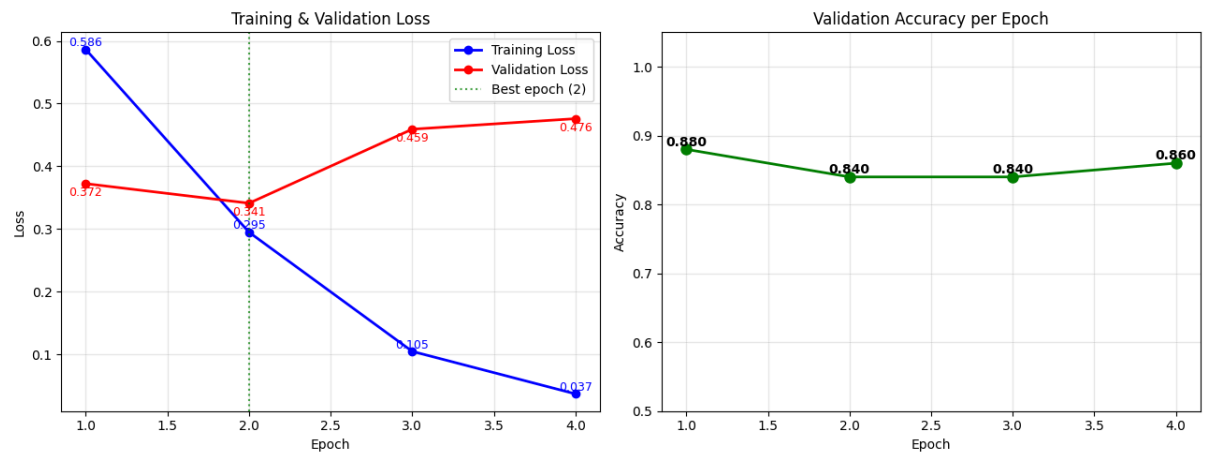


Figure 8: BERT training curves

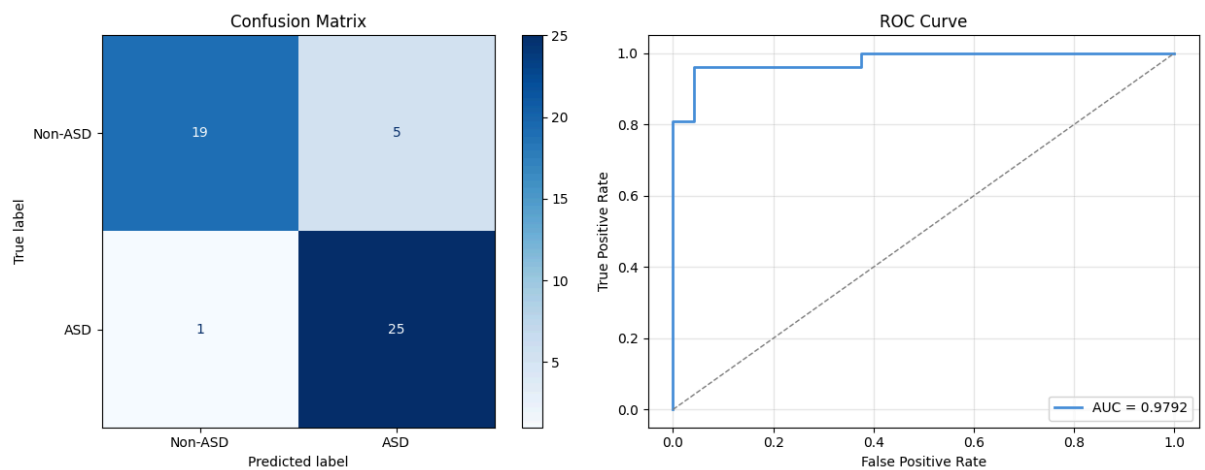


Figure 9: BERT ROC curve, confusion matrix

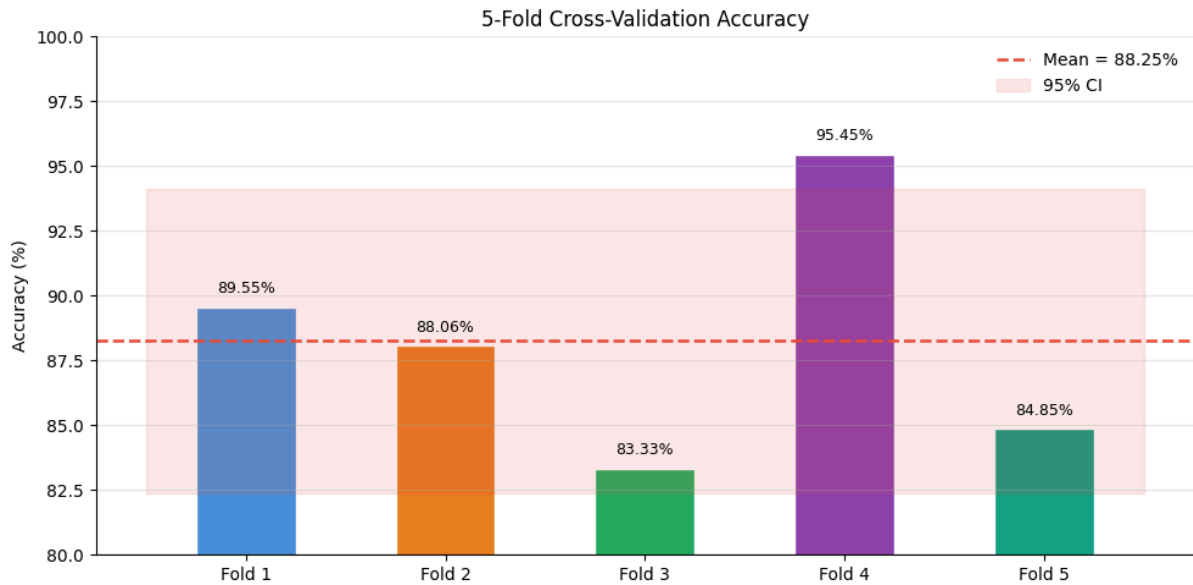


Figure 10: BERT 5-fold CV results

4.3 Performance Evaluation

4.3.1 System-Level Evaluation

The integration of two complementary modalities represents a key strength of the proposed system. The structured questionnaire modality provides highly accurate screening results when users are able to respond to predefined questions. In contrast, the free-text modality offers flexibility, allowing caregivers to describe behaviours in their own words.

This dual-modality design enhances accessibility by accommodating different user preferences and capabilities. It also improves the robustness of the system by providing alternative pathways for input, reducing reliance on any single interaction mode.

4.3.2 Comparison with Prior Work

The performance of the structured modality exceeds commonly reported accuracy ranges for similar datasets, which typically fall between 93% and 97%. This improvement can be attributed to rigorous preprocessing, proper handling of class imbalance, and the use of advanced ensemble learning techniques.

For text-based classification, the achieved performance represents a strong result given the limited size of the dataset. While direct comparisons with prior work are challenging due to differences in experimental setups, the results demonstrate the feasibility of applying transformer models to ASD-related text classification tasks.

4.3.3 Trait Taxonomy and Interpretability

An important aspect of the system is its ability to provide interpretable outputs through a behavioural trait taxonomy. By mapping model predictions to clinically meaningful categories, the system offers insights beyond simple binary classification. This enhances user understanding and supports informed decision-making.

The incorporation of a multi-level concern scoring system further improves usability by translating technical outputs into actionable information. This approach aligns with the goal of making AI systems accessible to non-specialist users.

4.4 Discussion

The results demonstrate that the proposed framework successfully achieves its primary objectives. The structured modality provides highly accurate screening capabilities, while the text-based modality enables flexible and natural interaction. Together, these components form a comprehensive system capable of supporting a wide range of users.

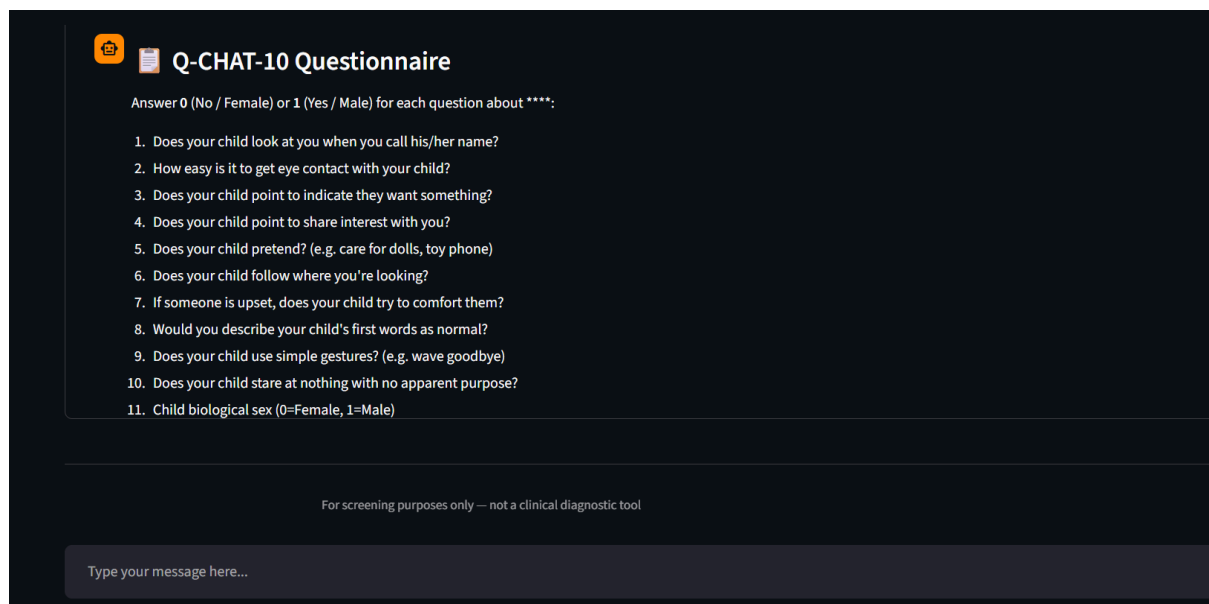
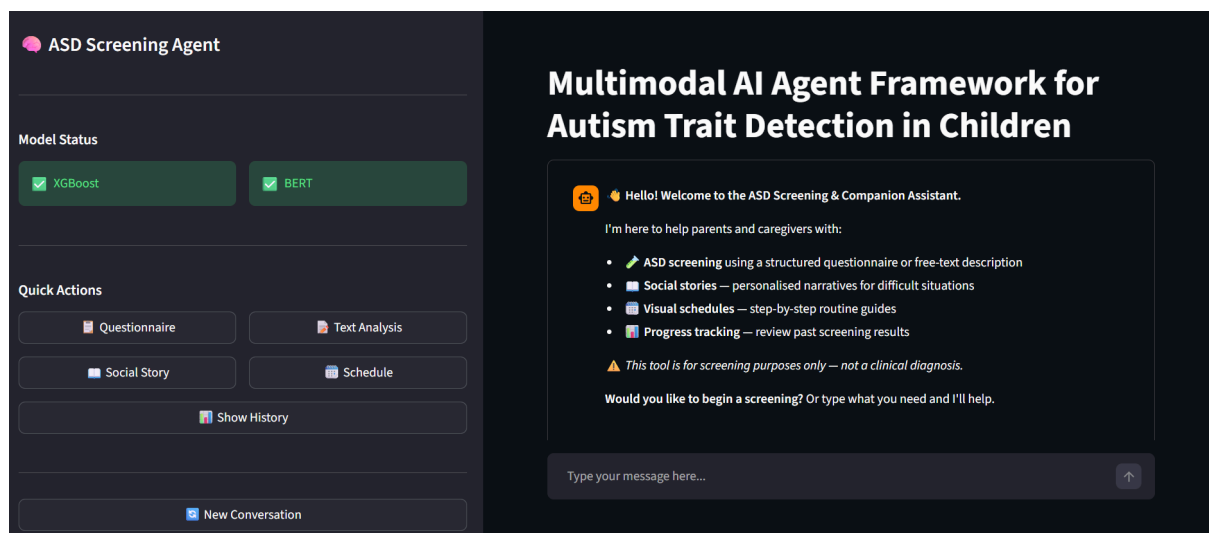
A key contribution of this work is the architectural design that separates machine learning outputs from language model interactions. This design choice directly addresses concerns regarding the reliability of generative AI systems in healthcare contexts, ensuring that critical decisions are based solely on validated models.

The integration of companion features, such as personalised content generation and semantic retrieval, extends the system's functionality beyond screening. This transforms the system into a continuous support tool, aligning with the long-term needs of caregivers and children with ASD.

Despite these strengths, several limitations remain, including dataset size constraints, lack of real-world validation, and language restrictions. These limitations highlight important directions for future research and development.

Chapter 5: Implementation

The proposed Multimodal AI Agent Framework for Autism Trait Detection in Children was implemented as an end-to-end system integrating machine learning models, a multi-agent orchestration layer, and an interactive web-based user interface. The implementation was structured in a modular manner to ensure scalability, maintainability, and clear separation of responsibilities between components.



 0,1,0,1,1,0,0,0,1,0,0

Questionnaire Analysis Complete

Field	Value
Assessment	Non-ASD
Confidence	92.2%
ASD probability	7.8%
Non-ASD probability	92.2%

 **Disclaimer:** This is a screening tool only, not a medical diagnosis. Always consult a qualified healthcare professional.

For screening purposes only — not a clinical diagnostic tool

 **Note:** Detailed behavioural trait analysis is available when using the **text description method**. The questionnaire provides a validated screening score.

You can ask me what the result means, what to do next, request a **social story** or **visual schedule**, type **another** for a new assessment, or **exit** to end.

 what is this result mean

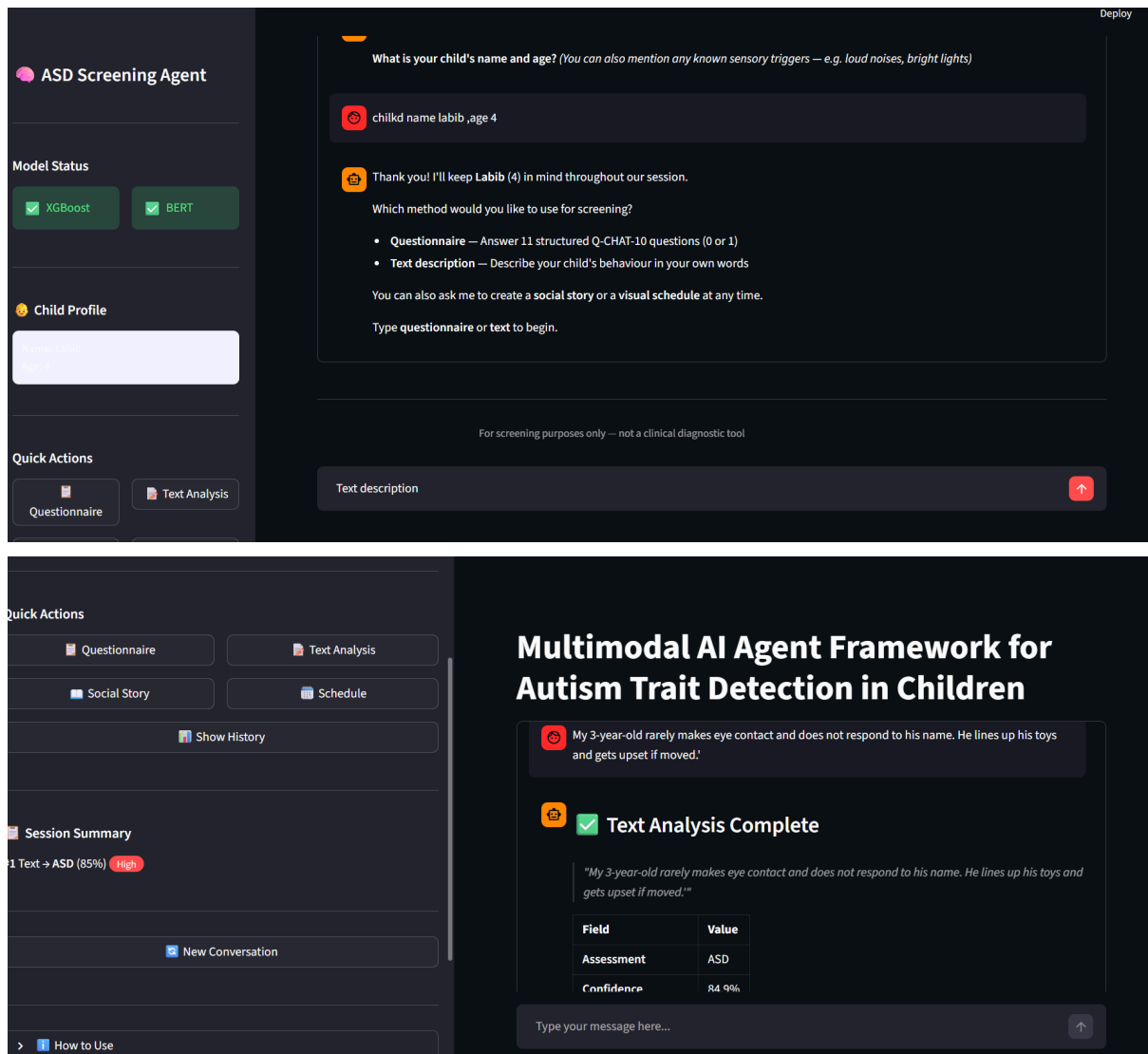
 I'm happy to help you understand the results, Sharafat's parent.

The assessment shows that the likelihood of Sharafat having Autism Spectrum Disorder (ASD) is **low**, with a confidence level of **92.2%**. This means that based on the answers you provided, Sharafat's behaviors and characteristics are not suggesting a high probability of ASD.

What does this mean for Sharafat?

At 4 years old, Sharafat is likely developing social, communication, and play skills that are typical for their age. The results suggest that Sharafat is probably interacting with others, using

For screening purposes only — not a clinical diagnostic tool



Chapter 6: Conclusion

6.1 Overall Contributions

This project presented the design and implementation of a multimodal AI agent framework for ASD trait detection in children. The system integrates structured and unstructured data processing within a unified architecture, enabling flexible and accessible screening.

The structured modality, based on XGBoost, achieved high accuracy and demonstrated strong generalisation across multiple evaluation settings. The text-based modality, implemented using a fine-tuned transformer model, achieved competitive performance despite operating under data-constrained conditions.

A multi-agent architecture was developed to manage system functionality, enabling modular design and efficient interaction handling. The implementation of an anti-hallucination boundary ensures that diagnostic outputs remain reliable, addressing a critical challenge in the deployment of AI systems in healthcare.

Additional contributions include the development of a behavioural taxonomy, a concern scoring mechanism, and the integration of supportive features such as social story generation and visual scheduling.

6.2 Limitations and Future Work

While the proposed system demonstrates strong potential, several limitations must be acknowledged. The size of the text dataset limits the performance and generalisability of the transformer model. Future work should focus on collecting larger and more diverse datasets, potentially through clinical collaboration.

The system is currently limited to English-language inputs, restricting accessibility for non-English-speaking populations. Extending the framework to support multiple languages represents an important area for future development.

Furthermore, the system has not yet been evaluated in real-world clinical settings. Conducting user studies and clinical validation trials will be essential for assessing practical usability and effectiveness.

Additional improvements could include the development of dedicated multi-label classifiers for behavioural traits, as well as the exploration of lightweight models for deployment in resource-constrained environments.

6.3 Final Remarks

The proposed framework demonstrates the potential of combining machine learning, natural language processing, and agent-based system design to address real-world healthcare challenges. By prioritising accessibility, reliability, and usability, this work contributes to the development of scalable tools for early ASD screening and support.

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Appendix

Program Outcomes

PO1 (Engineering Knowledge):

This project applies mathematical, scientific, and engineering principles to develop a deep learning–based lung histopathology classification system. Multiple CNN architectures (ResNet50, DenseNet121, EfficientNetB0, MobileNetV2, MobileNetV3Small) and ensemble techniques such as weighted voting, stacking, and boosting demonstrate strong grounding in machine learning, computer vision, and engineering fundamentals.

PO2 (Problem Analysis):

The project identifies key challenges in histopathology images, such as stain variability, magnification differences, class imbalance, and patient-level generalization. The introduction of patient-wise splitting and ensemble fusion techniques reflects systematic problem analysis and solution formulation.

PO3 (Design/Development of Solutions):

A dynamic hybrid ensemble system is designed, allowing flexible combinations of CNN models without retraining. Weighted voting, stacking, and boosting pipelines provide alternative diagnostic mechanisms and optimize prediction reliability.

PO4 (Investigation):

The project conducts extensive experimentation across five CNN backbones, three boosting models, and multiple ensemble strategies. It evaluates image-level and patient-level performance, validating findings with controlled experiments and comparative analysis.

PO5 (Modern Tool Usage):

Modern tools and frameworks such as TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, XGBoost, LightGBM, CatBoost, and Google Colab GPUs are used for model development, training, and evaluation. FastAPI is used for deployment testing.

PO6 (Engineer and Society):

The system contributes to healthcare by assisting in early lung cancer detection and supporting clinical decision-making. The dataset contains no personal patient information, ensuring ethical compliance and protecting privacy.

PO7 (Environment and Sustainability):

Lightweight CNN backbones and efficient ensemble strategies are explored to reduce computational cost, making the system sustainable for real-world medical environments with limited hardware resources.

PO8 (Ethics):

Ethical standards regarding dataset usage, privacy preservation, and transparent reporting are maintained. All dataset sources are publicly available and anonymized.

PO9 (Individual and Teamwork):

The project combines teamwork in experimentation, documentation, coding, and analysis. Group members collaborated in structured workflows, version control, and peer review.

PO10 (Communication):

The research communicates results clearly through graphs, tables, and structured descriptions. The FastAPI-based demo enables intuitive communication of system behavior to technical and non-technical audiences.

PO11 (Project Management and Finance):

The project was executed within a planned timeline using free or academic tools (Google Colab, Python libraries). Efficient use of cloud resources and proper task distribution ensured cost-effective completion.

PO12 (Life Long Learning):

The project reflects continuous learning in deep learning, ensemble modeling, deployment frameworks, and medical imaging. It encourages future extension into clinical applications and research advancements.

Mapping of Course and Program Outcomes

Addressing of COs, Knowledge Profile (K), and Complex Engineering Problem(EP):

Program Outcome (PO)	PO Description	Project Evidence
PO1	Engineering Knowledge: Apply knowledge of mathematics, science, and engineering fundamentals	Application of statistical methods (chi-square test, point-biserial correlation), probability metrics (AUC-ROC, Wilson confidence interval), gradient boosting principles using XGBoost [10], and transformer-based attention mechanisms in BERT [6].
PO2	Problem Analysis: Identify and analyse complex engineering problems	Comprehensive analysis of ASD screening challenges including modality limitations, data scarcity, and LLM hallucination risks [2]; identification of the need for a dual-modality system combining structured and free-text inputs.
PO3	Design/Development of Solutions: Design systems meeting specified needs	Development of a multimodal AI framework integrating XGBoost and BERT classifiers; design of a LangGraph-based multi-agent architecture with FSM routing and an anti-hallucination boundary ensuring reliable outputs.
PO4	Investigation: Conduct experiments and interpret data	Implementation of comparative experiments across multiple ML models; use of 5-fold cross-validation, ROC analysis, and statistical evaluation; augmentation experiments based on EDA techniques [4] to

		validate improvements in text classification.
PO5	Modern Tool Usage: Use modern engineering tools effectively	Utilisation of advanced tools including XGBoost [10], BERT via HuggingFace Transformers [6], FAISS for semantic retrieval [11], Sentence-BERT for embeddings [12], and Streamlit for interface development.
PO6	The Engineer and Society: Assess societal and health impacts	Design of the system as a screening (not diagnostic) tool aligned with clinical responsibility; incorporation of anti-hallucination safeguards inspired by risks identified in medical LLM usage [2]; focus on accessibility for low-resource settings.
PO7	Environment and Sustainability: Understand sustainable impact	Development of a scalable, low-cost digital screening system with near-zero marginal cost per use; potential deployment in underserved regions lacking access to ASD specialists.
PO8	Ethics: Apply ethical principles in engineering practice	Ethical exclusion of sensitive attributes such as ethnicity; use of anonymised public datasets [7]; clear distinction between screening and diagnosis; responsible AI design to avoid misleading outputs.
PO9	Individual and Team Work: Work effectively in teams and multidisciplinary settings	Execution of the project as an individual capstone integrating knowledge from machine learning, NLP, software engineering, and clinical research domains.
PO10	Communication: Communicate effectively with diverse audiences	Presentation of results through structured report, tables, and visualisations; development of a Streamlit-based conversational

		interface enabling communication with non-technical caregivers.
PO11	Project Management and Finance: Apply management principles	Efficient project execution using cloud-based tools (Google Colab) and open-source frameworks; inclusion of a business model covering deployment, cost considerations, and scalability.
PO12	Life-long Learning: Engage in continuous learning	Application of recent advancements in NLP, machine learning, and AI in healthcare [2], [5]; identification of future research directions including multilingual models and clinical validation.